

Scale, Completeness, and Conjugate Geometry of Landmark Metrics: Classification and Nondegenerate Designs

Run `fa_banach_001`, lane 12

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Status

This packet gives a structural theorem suite for spatial scale covariance, geodesic completeness, and conjugate geometry of landmark metrics.

1. For an arbitrary positive-definite landmark cometric $G(q)$, exact dilation covariance of the geodesic flow is equivalent to one power law $G(\lambda q) = \lambda^\alpha G(q)$.
2. A fixed continuous stationary kernel can obey this law only by becoming constant and hence degenerate for two or more landmarks.
3. Nondegenerate exactly covariant metrics exist as soon as the bandwidth is allowed to follow either the configuration or a global scale coordinate. All jointly covariant global-scale kernel families are classified by

$$\kappa_s(x) = s^\alpha \phi(x/s).$$

4. Homogeneity itself forces a sharp completeness obstruction: only the isometric exponent $\alpha = 2$ can put both total collapse and spatial infinity at infinite distance.
5. For every strictly positive-definite matrix profile whose RKHS embeds in C_b^1 , the adaptive RMS-normalized metric is geodesically complete for every N, d exactly at $\alpha = 2$. The Gaussian is only one instance of this admissible-profile theorem.
6. More generally, an Osgood feature modulus ω , with $\int_0^1 dt/\omega(t) = \infty$, is sufficient for all- N completeness at $\alpha = 2$. In fact only the feature gap in the instantaneous pair-separation direction needs an Osgood bound. For scalar profiles with a monotone kernel gap this integral condition is also necessary when $N \geq 3$, whereas $N = 2$ is always complete at $\alpha = 2$. In particular, the strict family $e^{-|z|^\beta}$ is complete for $N \geq 3$ exactly at $(\alpha, \beta) = (2, 2)$.
7. For every continuous strict stationary matrix profile, the entire adaptive two-landmark metric is complete exactly at $\alpha = 2$. For a matrix-valued TRI profile with locally monotone longitudinal gap $\delta_{\parallel}$, the all- N criterion is exact: for $N \geq 3$, completeness holds exactly when $\alpha = 2$ and $\int_0 dr / \sqrt{\delta_{\parallel}(r)} = \infty$. The tangential gap does not enter.
8. For an arbitrary, possibly irregular non-TRI matrix profile and every N , completeness has an exact criterion: quotient by translations and dilations and test completeness of an explicit Schur-complement metric on the compact normalized pre-shape sphere with its collision diagonals deleted. This is the intrinsic matrix collision-end object for which the Osgood and matrix-TRI tests are computable certificates.

9. For every matrix profile, including profiles with an odd skew part, the full adaptive two-landmark metric has an explicit center–relative block formula. For even profiles its fixed-center sector is an exact Kaluza–Klein metric over S^{d-1} , reducing relative geodesics and Jacobi fields to compact magnetic mechanics.
10. For every scalar radial profile, the entire two-landmark manifold at $\alpha = 2$ is exactly a scaled product $\mathbb{H}^{d+1} \times S^{d-1}$. Its distance, curvature, injectivity radius, cut locus, and full ambient conjugate locus are therefore explicit.
11. For an arbitrary matrix-valued TRI profile, the adaptive two-landmark fixed-center sector is an exact line–sphere product with independently determined radial and spherical scales.

The exponent $\alpha = 2$ makes dilation an actual isometry, not merely a homothety preserving geodesics.

The source conclusion of Micheli and Glaunès [1] is a research program, not one formal conjecture. The packet now completes the exact-scale subprogram, the all- N completeness problem for the full admissible adaptive-profile class, and the entire two-landmark ambient geometry for scalar profiles. It also gives an exact normalized-shape criterion for every arbitrary matrix profile, extends completeness to Osgood feature moduli, settles the exact rough/smooth phase diagram for power-exponential adaptive profiles, proves universal two-landmark completeness for matrix profiles, and settles all- N adaptive completeness for matrix-valued TRI profiles with a locally monotone longitudinal gap. Application-optimal kernel selection and a profile-independent closed formula for the full $N \geq 3$ conjugate locus remain outside this theorem suite; arbitrary-matrix completeness is now exactly reduced to a compact collision-end metric, while scalar and matrix-TRI monotone-gap completeness are explicitly classified.

Several literature corrections are essential. Niethammer and Vialard [2] already proved a broader nonexistence theorem for scale-invariant geodesic flow of smooth right-invariant LDDMM metrics. Thus the fixed-kernel obstruction is not claimed as a new literature resolution. Habermann–Preston–Sommer [6] completely characterize geodesic completeness for fixed scalar radial landmark kernels, and Micheli–Michor–Mumford [7] developed scalar landmark curvature and conjugate-point mechanisms. The candidate new portion is therefore confined to the homogeneous-cometric classification, the adaptive/global-scale designs, the admissible-profile completeness theorem, the exact $\mathbb{H}^{d+1} \times S^{d-1}$ two-landmark geometry, the matrix-TRI relative reductions, the configuration-normalized Osgood criterion and its directional refinement, the universal matrix two-landmark theorem, the exact scalar and matrix-TRI completeness criteria, the power-exponential phase diagrams, the arbitrary-matrix normalized-shape classification, the general two-landmark block and magnetic reductions, and the tail-decoupling estimate.

Source Direction and Prior Result

For a fixed translation-invariant matrix kernel $K(x, y) = \kappa(x - y)$, the source landmark Hamiltonian is

$$H(p, q) = \frac{1}{2} \sum_{a,b=1}^N p_a^\top \kappa(q^a - q^b) p_b, \quad \dot{q}^a = \sum_{b=1}^N \kappa(q^a - q^b) p_b.$$

The source conclusion observes that all examples contain a length constant, so their landmark and diffeomorphism dynamics are not scale invariant:

Figure 12: (A) Detail of the exponential map and (B) a landmark trajectory in the divergence-free case.

6 Conclusions

In this paper we have provided a thorough and systematic study of matrix-valued definite positive kernels that induce translation- and rotation-invariant metrics in the corresponding Reproducing Kernel Hilbert Spaces. We have conducted a novel analysis of such kernels in the Fourier domain: here the classification of spaces of curl-free and divergence free vector fields is very natural, in that each corresponds to one of the two coefficients of the Fourier transform of the kernel being equal to zero. This allows one to perform the Hodge decomposition in RKHS via the computation of Hankel transforms (which may be done analytically or numerically). We have also devised a method for building curl-free and divergence-free kernels via an appropriate differentiation of scalar kernels; we have also proven that this constructive procedure is general, in that any TRI kernel (more precisely, the terms of its Hodge decomposition) may be obtained this way. Such method may thus be used to design a large class of positive kernels. We have given a summary of the application of the LDDMM approach to the interpolation of vector fields and the matching of landmark points, together with the differential-geometric interpretation and the derivation of the corresponding Hamiltonian. Finally, we have given numerical examples of the dynamics of landmark points and the corresponding flow of diffeomorphisms for different choices of TRI kernels.

Future work should focus on the design of kernels that are appropriate for different applications, especially in computational anatomy and related fields. For example, the shape of geodesics should be studied in relation to the choice of the kernel; in particular, note that all the kernels that we have considered in the examples have a length constant (e.g. the parameter $\sigma > 0$ for Bessel-type kernels), so that the induced dynamics in the space of landmarks (and, for that matter, in the group of diffeomorphisms itself) is not scale invariant; furthermore, the regularity of the kernels at zero and the heaviness of their tails are also crucial in determining both the qualitative dynamics and the complexity of numerical implementations. Last, but not least, the visualization of the exponential map in the previous section (especially in the case of divergence-free TRI kernels) reveals the presence of several conjugate points, i.e. landmark configurations that are connected by distinct geodesics; the study of the dynamics of $\mathcal{L}^N(\mathbb{R}^d)$ induced by the action of diffeomorphism groups should be furthered, and the investigation of the differential geometry of the space of landmarks that was started in [38] in the case of scalar kernels, which is useful for the ultimate goal of performing statistics on such manifold, should be extended to the case of generic TRI kernels.

The bounded follow-up search located the following directly relevant prior result. Niethammer–Vialard’s Theorem 3.2 (printed as Theorem 32) rules out scale-invariant geodesic flow for every smooth right-invariant metric on the diffeomorphism group when the symmetry group contains translations, rotations, and dilations [2].

rotations. The LDDMM framework is built on a group with a right-invariant distance that acts on the left on the space of shapes. It then induces a Riemannian metric on each orbit. A priori, requiring scale invariance for the induced metric is less demanding than asking it on the group itself. However, let us first explore the case of scale invariance on the group:

Definition 1 *The geodesic flow is invariant under G_0 if for any geodesics $t \mapsto \phi(t)$ the curve $t \mapsto g_0 \phi(t) g_0^{-1}$ is also a geodesic.*

Remark 31 *This definition implies the following fact: if (q_0, q_1) are two objects connected by the geodesic $q(t)$ then $g_0 \cdot q(t)$ is a geodesic connecting $(g_0 \cdot q_0, g_0 \cdot q_1)$.*

Theorem 32 *There does not exist any smooth right-invariant metric on the group of diffeomorphisms of $\mathbb{R}^n$ for which the geodesic flow is invariant under G_0 that contains rotations, translations and the scaling transformations.*

Proof. Let us consider a geodesic path $\phi(t)$ whose vector field is denoted by $v(t)$. By Ad invariance the vector field associated with $g_0 \phi(t) g_0^{-1}$ is $u(t) := \text{Ad}_{g_0}(v(t))$. In addition to that, we know that $\phi(t)$ is a geodesic on the group if and only if it satisfies the Euler-Poincaré equation which is

$$\dot{m}(t) + \text{ad}_{v(t)}^* m(t) = 0, \quad (7)$$

where $m(t) = Lv(t)$ (or $v(t) = Km(t)$). Equivalently, we have

$$\dot{v}(t) + K \text{ad}_{v(t)}^* Lv(t) = 0, \quad \text{Ad}_{g_0}(\dot{v}(t)) + K \text{ad}_{u(t)}^* Lu(t) = 0.$$

This implies that

$$\text{Ad}_{g_0}^{-1}(K \text{ad}_{\text{Ad}_{g_0} v(t)}^* L \text{Ad}_{g_0} v(t)) = K \text{ad}_{v(t)}^* Lv(t), \quad (8)$$

That theorem is broader at the diffeomorphism-group level. The elementary landmark argument below gives a sharp finite-dimensional classification, including the degenerate boundary, and then identifies two precise ways to leave the fixed-kernel setting.

Definitions and Proof Intuition

Let

$$\mathcal{Q}_N = \{q = (q^1, \dots, q^N) \in (\mathbb{R}^d)^N : q^a \neq q^b \text{ for } a \neq b\}.$$

A C^1 positive-definite cometric is a map $G : \mathcal{Q}_N \rightarrow \text{Sym}_{++}(Nd)$. Its geodesic Hamiltonian is

$$H(q, p) = \frac{1}{2} p^\top G(q) p. \quad (1)$$

We call its flow *exactly dilation covariant* if for every $\lambda > 0$ there are $A_\lambda, C_\lambda > 0$, independent of the solution, such that

$$Q(t) = \lambda q(C_\lambda t), \quad P(t) = A_\lambda p(C_\lambda t) \quad (2)$$

is again a solution whenever (q, p) is.

The proof has one central observation. Since $\dot{q} = G(q)p$, comparing the first Hamilton equation before and after dilation, for arbitrary p , forces

$$G(\lambda q) = h_\lambda G(q).$$

Composition of dilations makes h a continuous positive multiplicative character, hence $h_\lambda = \lambda^\alpha$. Conversely, differentiating this homogeneity law gives exactly the scaling needed in the momentum equation. Exact scale covariance is therefore equivalent to homogeneity of the cometric.

A fixed stationary kernel cannot be homogeneous without collapsing at its diagonal value. But if its bandwidth is chosen by a homogeneous configuration scale $\rho(q)$, the normalized differences

$(q^a - q^b)/\rho(q)$ do not change under dilation. This immediately produces nondegenerate homogeneous Gram matrices. The same idea with an independent global scale s yields a sharp classification of all covariant scale-indexed families.

The completeness proofs use the cometric directly. A covector such as $d\rho$ or the difference map $q \mapsto q^a - q^b$ has an easily bounded cometric norm. The pair bound is exactly the squared norm of a difference of RKHS feature maps, so ordinary C_b^1 -admissibility supplies the needed quadratic estimate for every matrix-valued profile. Dual Cauchy–Schwarz then turns these bounds into logarithmic lower bounds on the length required for total collapse, escape, or a pair collision.

The logarithm is not essential. If the feature difference has any nondecreasing modulus ω , the same argument controls the variation of the Osgood coordinate $\int dt/\omega(t)$. Divergence of that integral still makes every collision infinitely far away. Conversely, for a rough power-exponential profile and $N \geq 3$, let two landmarks collide symmetrically while the others remain fixed. In the antisymmetric feature direction the Gram matrix has a Schur complement $S(\varepsilon) \sim C\varepsilon^\beta$; coupling to all noncolliding features is only $O(\varepsilon^2)$. The path speed is therefore asymptotic to $C\varepsilon^{-\beta/2}$, which is integrable exactly when $\beta < 2$. This produces a finite-length collision and explains the sharp threshold.

For a matrix profile, controlling the whole feature-gap operator is stronger than necessary. The differential of the distance between a pair is the covector in its instantaneous separation direction, so only the quadratic form of the feature gap in that same direction enters. For a TRI profile this is exactly twice the longitudinal gap $\delta_{\parallel} = a - k_{\parallel}$. Collinear configurations preserve the longitudinal/transverse splitting, so the scalar Schur argument becomes an exact necessity proof in the longitudinal sector. The tangential gap is irrelevant to collision completeness.

For a completely irregular non-TRI profile, no single radial output direction diagonalizes all collision modes. The exact replacement is to remove the two isometric nuisance actions—translations and positive dilations. Every orbit has a unique centered, RMS-one representative on a compact pre-shape sphere. Orthogonally minimizing over the translation and dilation directions gives an explicit Schur-complement quotient metric. A finite-length landmark collision projects to a finite-length path to a deleted diagonal, and every such quotient path has a horizontal lift of the same length. Thus completeness of the original metric is exactly completeness of this compact collision-end metric.

For two landmarks normalization freezes the profile at separation two. For every matrix profile, compactness of the direction sphere makes the resulting normalized two-point Gram matrix uniformly positive, and the metric is uniformly equivalent to $\mathbb{H}^{d+1} \times S^{d-1}$; this already proves completeness. In the scalar case the center/relative Hamiltonian further diagonalizes, and a constant rescaling of the center coordinate converts the (c, r) -metric into the upper-half-space metric on $\mathbb{H}^{d+1}$, while the direction of the relative vector supplies a round S^{d-1} . Thus the full conjugate and cut geometry is a product calculation, not a numerical phenomenon.

Without evenness, a matrix profile can have an odd skew part that couples center and relative momenta. A direct canonical block calculation isolates that coupling. When the profile is even it disappears, and the fixed-center relative metric completes the square as a Kaluza–Klein metric over the direction sphere. Conservation of logarithmic-scale momentum then gives an exact magnetic Routh reduction. The TRI line–sphere product is the zero-connection, constant-coefficient specialization.

The General Classification

Theorem 1 (Homogeneous cometrics are exactly the dilation-covariant ones). *Let $G : \mathcal{Q}_N \rightarrow \text{Sym}_{++}(Nd)$ be C^1 . The Hamiltonian flow of (1) is exactly dilation covariant in the sense of (2) if*

and only if there is $\alpha \in \mathbb{R}$ such that

$$G(\lambda q) = \lambda^\alpha G(q) \quad (\lambda > 0, q \in \mathcal{Q}_N). \quad (3)$$

In that case the admissible rescalings satisfy

$$A_\lambda = C_\lambda \lambda^{1-\alpha}.$$

If $g = G^{-1}$ is the corresponding Riemannian metric and $D_\lambda(q) = \lambda q$, then

$$D_\lambda^* g = \lambda^{2-\alpha} g. \quad (4)$$

Thus every α gives exact covariance of affinely parametrized geodesics, while $\alpha = 2$ makes every D_λ an isometry.

Proof. Suppose first that the flow is covariant. Hamilton's first equation applied to (2) gives, at $t = 0$,

$$\lambda C_\lambda G(q)p = A_\lambda G(\lambda q)p.$$

The initial momentum is arbitrary, so

$$G(\lambda q) = h_\lambda G(q), \quad h_\lambda := \frac{\lambda C_\lambda}{A_\lambda} > 0.$$

Applying this identity successively with λ and μ and using that $G(q)$ is invertible gives $h_{\lambda\mu} = h_\lambda h_\mu$. For any fixed q and nonzero u ,

$$h_\lambda = \frac{u^\top G(\lambda q)u}{u^\top G(q)u},$$

so h is continuous. The continuous positive multiplicative characters of $(0, \infty)$ are $h_\lambda = \lambda^\alpha$. This proves (3) and the displayed relation between A_λ and C_λ .

Conversely, differentiating (3) in q gives

$$DG(\lambda q) = \lambda^{\alpha-1} DG(q).$$

Choose any $C_\lambda > 0$ and put $A_\lambda = C_\lambda \lambda^{1-\alpha}$. The transformed first Hamilton equation scales by

$$A_\lambda \lambda^\alpha = \lambda C_\lambda.$$

The right side of the transformed momentum equation scales by

$$A_\lambda^2 \lambda^{\alpha-1} = A_\lambda C_\lambda,$$

which is exactly the factor obtained by differentiating $P(t) = A_\lambda p(C_\lambda t)$. Hence both Hamilton equations hold.

Finally, $G(\lambda q) = \lambda^\alpha G(q)$ implies $g_{\lambda q} = \lambda^{-\alpha} g_q$. Since $dD_\lambda = \lambda I$, pulling back the metric gives (4). A constant-factor homothety preserves the Levi-Civita connection, hence affinely parametrized geodesics. $\square$

Why Fixed Stationary Kernels Must Degenerate

Corollary 1 (Sharp fixed-kernel obstruction). *Let $K(x, y) = \kappa(x - y)$ be a continuous positive-definite matrix-valued kernel. If its N -landmark block Gram cometric is nondegenerate for $N \geq 2$, its geodesic flow is not exactly dilation covariant. If degeneracy is allowed, exact covariance forces $\kappa(x) = \kappa(0)$; conversely every constant positive-semidefinite matrix kernel is covariant and degenerate.*

Proof. In the nondegenerate case, Theorem 1 gives

$$\kappa(\lambda(q^a - q^b)) = \lambda^\alpha \kappa(q^a - q^b).$$

The diagonal blocks give $\kappa(0) = \lambda^\alpha \kappa(0)$. A nonzero positive-definite stationary kernel has $\kappa(0) \neq 0$: the kernel Cauchy–Schwarz inequality

$$|u^\top \kappa(x) v|^2 \leq (u^\top \kappa(0) u)(v^\top \kappa(0) v)$$

shows that $\kappa(0) = 0$ would imply $\kappa \equiv 0$. Thus $\alpha = 0$, and continuity as $\lambda \downarrow 0$ gives $\kappa(x) = \kappa(0)$.

The same conclusion holds without assuming a nondegenerate Gram matrix. Indeed, use two landmarks with initial positions $(x, 0)$ and momenta $(0, v)$. Comparing the first Hamilton equation under (2) directly gives

$$\kappa(\lambda x) = h_\lambda \kappa(x).$$

Positive definiteness and nonzeroness give $\kappa(0) \neq 0$. Letting $x \rightarrow 0$ in the displayed identity forces $h_\lambda = 1$, and continuity again makes κ constant.

The N -landmark Gram matrix is then

$$(\mathbf{1}_N \mathbf{1}_N^\top) \otimes \kappa(0),$$

whose rank is at most $d < Nd$. Conversely, a constant kernel has $\dot{p}_a = 0$ and $\dot{q}^a = \kappa(0) \sum_b p_b$, so $Q = \lambda q$, $P = \lambda p$ is again a solution. $\square$

Remark 1. *In the source’s stronger assumption $V \hookrightarrow C_0^1(\mathbb{R}^d, \mathbb{R}^d)$, a constant kernel section belongs to C_0^1 only when it is zero. Thus the only fixed exactly covariant kernel in that class is the zero kernel.*

A Nondegenerate Adaptive Kernel on Landmark Space

Theorem 2 (Configuration-adaptive exact scale covariance). *Let $\rho : \mathcal{Q}_N \rightarrow (0, \infty)$ be C^2 and homogeneous of degree one,*

$$\rho(\lambda q) = \lambda \rho(q).$$

Let $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ be a C^2 profile for which $(x, y) \mapsto \phi(x - y)$ is a strictly positive-definite matrix kernel. For any $\alpha \in \mathbb{R}$, define the block cometric

$$G_\alpha(q)_{ab} = \rho(q)^\alpha \phi\left(\frac{q^a - q^b}{\rho(q)}\right). \quad (5)$$

Then $G_\alpha(q)$ is positive definite on $\mathcal{Q}_N$, and its geodesic flow is exactly dilation covariant. The induced metric satisfies

$$D_\lambda^* g_\alpha = \lambda^{2-\alpha} g_\alpha.$$

In particular, $\alpha = 2$ gives a nondegenerate scale-invariant Riemannian metric on landmark space.

Proof. The normalized points $q^a/\rho(q)$ remain pairwise distinct, so strict positive definiteness of ϕ makes the block matrix in (5) positive definite. Since

$$\frac{\lambda q^a - \lambda q^b}{\rho(\lambda q)} = \frac{q^a - q^b}{\rho(q)},$$

we have

$$G_\alpha(\lambda q) = \lambda^\alpha G_\alpha(q).$$

The conclusion follows from Theorem 1. $\square$

Corollary 2 (Explicit RMS–Gaussian metric). *Let*

$$\bar{q} = \frac{1}{N} \sum_{a=1}^N q^a, \quad \rho(q) = \left(\frac{1}{N} \sum_{a=1}^N |q^a - \bar{q}|^2 \right)^{1/2},$$

and $\phi(x) = e^{-|x|^2} I_d$. Formula (5) gives a smooth, nondegenerate landmark metric invariant under translations, rotations, and permutations. At $\alpha = 2$, it is also invariant under all spatial dilations.

Proof. Pairwise distinct landmarks are not all equal, so $\rho(q) > 0$ on $\mathcal{Q}_N$; the displayed ρ is smooth there and has all the stated symmetries. The scalar Gaussian is strictly positive definite. One direct proof uses its positive Fourier density:

$$\sum_{a,b} c_a^\top c_b e^{-|x_a - x_b|^2} = C_d \int_{\mathbb{R}^d} e^{-|\xi|^2/4} \left| \sum_a e^{i\xi \cdot x_a} c_a \right|^2 d\xi.$$

For distinct x_a , the exponential polynomial on the right cannot vanish identically unless all $c_a = 0$. Apply Theorem 2. $\square$

Remark 2 (The exact price of the construction). *The cometric in (5) does not arise from one fixed ambient stationary RKHS kernel: its bandwidth depends on the whole landmark configuration. This is precisely the assumption that Corollary 1 shows must be abandoned. For each fixed value of ρ , however, the profile is an ordinary smooth positive-definite kernel.*

A Dynamical Global Scale and Its Complete Kernel Law

The adaptive construction can be replaced by an independent global scale coordinate. This gives a useful classification rather than just an example.

Theorem 3 (Classification of global-scale kernel Hamiltonians). *Let $(\kappa_s)_{s>0}$ be a jointly C^2 , nonzero family of translation-invariant positive-definite matrix kernels, and let $b : (0, \infty) \rightarrow (0, \infty)$ be C^1 . On the augmented phase space with scale s and conjugate momentum π , consider*

$$\mathcal{H}(q, s; p, \pi) = \frac{1}{2} \sum_{a,b} p_a^\top \kappa_s(q^a - q^b) p_b + \frac{1}{2} b(s) \pi^2. \quad (6)$$

Say that the flow is jointly dilation covariant if, for every $\lambda > 0$, there are $A_\lambda, C_\lambda > 0$ such that every solution maps to

$$Q(t) = \lambda q(C_\lambda t), \quad S(t) = \lambda s(C_\lambda t), \quad P(t) = A_\lambda p(C_\lambda t), \quad \Pi(t) = A_\lambda \pi(C_\lambda t).$$

This holds if and only if there are $\alpha \in \mathbb{R}$, $\beta > 0$, and a positive-definite profile ϕ such that

$$\boxed{\kappa_s(x) = s^\alpha \phi(x/s), \quad b(s) = \beta s^\alpha.} \quad (7)$$

Then $A_\lambda = C_\lambda \lambda^{1-\alpha}$. If ϕ is strictly positive definite, $\mathcal{H}$ is a nondegenerate cometric on $\mathcal{Q}_N \times (0, \infty)$, and the augmented metric obeys

$$D_\lambda^* g = \lambda^{2-\alpha} g, \quad D_\lambda(q, s) = (\lambda q, \lambda s).$$

Proof. For necessity, use two landmarks with positions $(x, 0)$ and momenta $(0, v)$. Comparing the first q -equation before and after dilation gives

$$\kappa_{\lambda s}(\lambda x) = h_\lambda \kappa_s(x), \quad h_\lambda = \frac{\lambda C_\lambda}{A_\lambda}.$$

The scale equation is $\dot{s} = b(s)\pi$. Its transformed version gives

$$b(\lambda s) = h_\lambda b(s).$$

Since $b > 0$, composition implies $h_{\lambda\mu} = h_\lambda h_\mu$, and $h_\lambda = b(\lambda)/b(1)$ is continuous. Hence $h_\lambda = \lambda^\alpha$. Taking $s = 1$, then rescaling by s , yields (7) with $\phi = \kappa_1$ and $\beta = b(1)$.

For sufficiency, joint homogeneity and differentiation give

$$\begin{aligned} \kappa_{\lambda s}(\lambda x) &= \lambda^\alpha \kappa_s(x), \\ D_x \kappa_{\lambda s}(\lambda x) &= \lambda^{\alpha-1} D_x \kappa_s(x), \\ \partial_s \kappa_{\lambda s}(\lambda x) &= \lambda^{\alpha-1} \partial_s \kappa_s(x), \end{aligned}$$

and $b'(\lambda s) = \lambda^{\alpha-1} b'(s)$. With $A_\lambda = C_\lambda \lambda^{1-\alpha}$, the transformed q - and s -equations acquire the factor λC_λ ; the transformed p - and π -equations acquire

$$A_\lambda^2 \lambda^{\alpha-1} = A_\lambda C_\lambda.$$

These are exactly the factors obtained by differentiating the transformed variables, proving all four Hamilton equations.

The augmented cometric matrix is $\text{diag}(G_s(q), b(s))$, which is positive definite when ϕ is strictly positive definite. It scales by λ^α ; inversion and $dD_\lambda = \lambda I$ give the metric homothety law. $\square$

Corollary 3 (Frozen-scale family). *For a family (7) with s treated as an external parameter, every trajectory for s maps to a trajectory for λs by*

$$Q(t) = \lambda q(t), \quad P(t) = \lambda^{1-\alpha} p(t).$$

Thus a scale family restores exact covariance without making the scale dynamical.

Completeness Selects the Isometric Exponent

The covariance classification has an immediate global consequence that is independent of any kernel ansatz.

Theorem 4 (Radial completeness obstruction). *Let G be a positive-definite cometric on a dilation cone and suppose $G(rq) = r^\alpha G(q)$. If the Riemannian metric $g = G^{-1}$ is geodesically complete, then $\alpha = 2$. More precisely, $\alpha < 2$ puts the cone apex at finite distance along every dilation ray, whereas $\alpha > 2$ puts spatial infinity at finite distance.*

Proof. Fix u in the cone and use the curve $q(r) = ru$. Homogeneity gives $g_{ru} = r^{-\alpha}g_u$, so its length on an interval (a, b) is

$$\sqrt{u^\top g_u u} \int_a^b r^{-\alpha/2} dr.$$

The integral is finite at zero exactly when $\alpha < 2$, and finite at infinity exactly when $\alpha > 2$. Either finite-length exit makes the metric, hence by Hopf–Rinow the geodesic flow, incomplete. $\square$

The converse is false for an arbitrary degree-two cometric because a shape collision at fixed scale can still be finitely far away. It becomes true for the entire adaptive class generated by an admissible stationary RKHS profile.

Theorem 5 (Sharp completeness for admissible adaptive profiles). *Let $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ be C^2 , and suppose $(x, y) \mapsto \Phi(x - y)$ is a strictly positive-definite matrix kernel. Assume that for some $L > 0$,*

$$0 \preccurlyeq 2\Phi(0) - \Phi(z) - \Phi(-z) \preccurlyeq L^2|z|^2 I_d \quad (z \in \mathbb{R}^d). \quad (8)$$

For $N \geq 2$, put

$$\rho(q)^2 = \frac{1}{N} \sum_a |q^a - \bar{q}|^2, \quad G^{(\alpha)}(q)_{ab} = \rho(q)^\alpha \Phi\left(\frac{q^a - q^b}{\rho(q)}\right).$$

On every connected component of $\mathcal{Q}_N$, the induced Riemannian metric is geodesically complete if and only if $\alpha = 2$.

Condition (8) holds in particular when Φ is the profile of a stationary vector-valued RKHS V continuously embedded in $C_b^1(\mathbb{R}^d, \mathbb{R}^d)$.

Proof. Strict positive definiteness makes $G^{(\alpha)}(q)$ positive definite: the normalized landmarks remain pairwise distinct. For $\alpha \neq 2$, Theorem 4 supplies a finite-length dilation ray. It remains to prove completeness for $\alpha = 2$; write $G = G^{(2)}$ and $T = \text{tr } \Phi(0)$.

For a tangent vector v and covector ℓ , use

$$|v|_g^2 = v^\top G^{-1} v, \quad |\ell|_G^2 = \ell G \ell^\top, \quad |\ell(v)| \leq |\ell|_G |v|_g.$$

The normalized block Gram matrix is positive semidefinite and has trace NT . Hence

$$\|G(q)\|_{\text{op}} \leq NT \rho(q)^2. \quad (9)$$

Moreover

$$\nabla_{q^a} \rho = \frac{q^a - \bar{q}}{N\rho}, \quad |\nabla \rho|^2 = \frac{1}{N}.$$

Equation (9) gives

$$|d\rho|_G^2 \leq T\rho^2, \quad \left| \frac{d}{dt} \log \rho \right| \leq \sqrt{T} |\dot{q}|_g. \quad (10)$$

Thus both total collapse and scale escape require infinite length.

For a pair $a \neq b$, let $L_{ab}q = q^a - q^b$ and $r_{ab} = |q^a - q^b|$. Direct block subtraction and (8) give

$$\begin{aligned} L_{ab} G L_{ab}^\top &= \rho^2 \left[2\Phi(0) - \Phi\left(\frac{q^a - q^b}{\rho}\right) - \Phi\left(\frac{q^b - q^a}{\rho}\right) \right] \\ &\preccurlyeq L^2 r_{ab}^2 I_d. \end{aligned}$$

Applying dual Cauchy–Schwarz to dr_{ab} yields

$$\left| \frac{d}{dt} \log r_{ab} \right| \leq L |\dot{q}|_g. \quad (11)$$

Every pair collision is therefore infinitely far away.

A finite-length curve has ρ bounded above by (10). Equation (9) implies

$$|\dot{q}|_{\text{Eucl}} \leq \sqrt{NT} \rho |\dot{q}|_g,$$

so the curve is Euclidean Cauchy. It has a finite Euclidean limit, and (11) keeps every limiting pair separation positive. The limit lies in $\mathcal{Q}_N$. This finite-length escape criterion proves metric completeness, and Hopf–Rinow gives geodesic completeness.

Finally let $F_x u = K(\cdot, x)u$ be the feature map of the stationary RKHS. The reproducing identity gives

$$\|F_x u - F_y u\|_V^2 = u^\top [2\Phi(0) - \Phi(x - y) - \Phi(y - x)]u.$$

If $\|v\|_{C_b^1} \leq L\|v\|_V$, then

$$\begin{aligned} \|F_x u - F_y u\|_V &= \sup_{\|v\|_V=1} |u^\top (v(x) - v(y))| \\ &\leq L|u| |x - y|, \end{aligned}$$

which is (8). □

Corollary 4 (Adaptive RMS–Gaussian family). *For*

$$\Phi(z) = e^{-|z|^2} I_d,$$

the adaptive family in Theorem 5 is complete for every N, d exactly at $\alpha = 2$.

Proof. Here

$$2\Phi(0) - \Phi(z) - \Phi(-z) = 2(1 - e^{-|z|^2})I_d \preccurlyeq 2|z|^2 I_d.$$

□

Theorem 6 (Osgood feature-modulus completeness). *Let $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ be continuous, C^2 away from the origin, and suppose $(x, y) \mapsto \Phi(x - y)$ is a strictly positive-definite matrix kernel. Suppose there is a positive nondecreasing function $\omega : (0, \infty) \rightarrow (0, \infty)$ such that*

$$0 \preccurlyeq 2\Phi(0) - \Phi(z) - \Phi(-z) \preccurlyeq \omega(|z|)^2 I_d \quad (z \neq 0), \quad \int_0^1 \frac{dt}{\omega(t)} = \infty. \quad (12)$$

Then, for every $N \geq 2$, the RMS-normalized adaptive family in Theorem 5 is geodesically complete on every component of $\mathcal{Q}_N$ if and only if $\alpha = 2$.

Proof. Only sufficiency at $\alpha = 2$ is new. The trace argument does not use regularity at the origin, so (9) and (10) remain valid. Consequently, along a curve of finite total length there are constants $0 < m \leq M < \infty$ such that

$$m \leq \rho(q(t)) \leq M.$$

For $r_{ab} = |q^a - q^b|$, direct block subtraction and (12) give

$$L_{ab} G L_{ab}^\top \preccurlyeq \rho^2 \omega(r_{ab}/\rho)^2 I_d.$$

Dual Cauchy–Schwarz and monotonicity of ω therefore imply

$$|\dot{r}_{ab}| \leq \rho\omega(r_{ab}/\rho)|\dot{q}|_g \leq M\omega(r_{ab}/m)|\dot{q}|_g. \quad (13)$$

If r_{ab} approached zero, the change of variable

$$\Psi(s) = \int_s^{s_0} \frac{du}{M\omega(u/m)}$$

would tend to infinity by (12), whereas (13) gives $|\frac{d}{dt}\Psi(r_{ab}(t))| \leq |\dot{q}|_g$. This is impossible for a finite-length curve. Thus every pair separation stays positive. The operator estimate (9) again makes the curve Euclidean Cauchy, with a finite, pairwise-distinct limit. Metric completeness and Hopf–Rinow finish the proof. Necessity of $\alpha = 2$ is Theorem 4. $\square$

Remark 3 (The Osgood extension is genuine). *For $0 < c \leq 2$, the local modulus*

$$\omega(t) = t(1 - \log t)^{c/2} \quad (0 < t \ll 1)$$

satisfies $\int_0 dt/\omega(t) = \infty$, but $\omega(t)/t \rightarrow \infty$; hence it is not covered by the quadratic feature estimate (8). Habermann–Preston–Sommer [6] use precisely the scalar kernel gap $\varphi(0) - \varphi(t) = t^2(1 - \log t)^c$, $1 < c \leq 2$, in their fixed-kernel completeness examples. Whenever such a profile is used in the configuration-normalized construction, Theorem 6 supplies adaptive completeness as well.

Theorem 7 (Directional Osgood completeness). *Let Φ satisfy the continuity, off-origin C^2 , and strict positive-definiteness hypotheses of Theorem 6. Suppose there is a positive nondecreasing ω such that, for every $t > 0$ and unit vector e ,*

$$e^\top [2\Phi(0) - \Phi(te) - \Phi(-te)]e \leq \omega(t)^2, \quad \int_0^1 \frac{dt}{\omega(t)} = \infty. \quad (14)$$

Then, for every $N \geq 2$, the RMS-normalized adaptive family is geodesically complete exactly when $\alpha = 2$.

Proof. Only sufficiency is new. Let $z = q^a - q^b = re$, where $r > 0$ and $|e| = 1$, and let L_{ab} be the pair-difference map. Direct block subtraction gives

$$L_{ab}GL_{ab}^\top = \rho^2 \left[2\Phi(0) - \Phi\left(\frac{r}{\rho}e\right) - \Phi\left(-\frac{r}{\rho}e\right) \right].$$

Since $dr = e^\top L_{ab}$, its squared cometric norm satisfies

$$|dr|_G^2 = e^\top L_{ab}GL_{ab}^\top e \leq \rho^2\omega(r/\rho)^2.$$

Along a finite-length curve the scale estimate (10) gives $0 < m \leq \rho \leq M < \infty$. Dual Cauchy–Schwarz therefore yields

$$|\dot{r}| \leq M\omega(r/m)|\dot{q}|_g.$$

The Osgood-coordinate argument in Theorem 6 now excludes every pair collision. The trace estimate (9) makes the curve Euclidean Cauchy, so its limit lies in $\mathcal{Q}_N$. Hopf–Rinow proves completeness. Theorem 4 excludes $\alpha \neq 2$. $\square$

Theorem 8 (Universal adaptive two-landmark matrix completeness). *Let $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ be continuous and C^2 away from the origin, and suppose its stationary matrix kernel is strictly positive definite. For $N = 2$, the RMS-normalized adaptive metric is geodesically complete exactly when $\alpha = 2$.*

Proof. Necessity is Theorem 4. Let $\alpha = 2$, and write

$$c = \frac{q^1 + q^2}{2}, \quad q^1 - q^2 = re, \quad r > 0, \quad |e| = 1.$$

Then $\rho = r/2$, and the normalized block Gram matrix is

$$K(e) = \begin{pmatrix} \Phi(0) & \Phi(2e) \\ \Phi(-2e) & \Phi(0) \end{pmatrix}.$$

It is positive definite for every e . Continuity and compactness of S^{d-1} give constants $0 < m_0 \leq M_0 < \infty$ with

$$m_0 I_{2d} \preceq K(e) \preceq M_0 I_{2d} \quad (e \in S^{d-1}).$$

Moreover

$$|dq|_{\text{Eucl}}^2 = 2|dc|^2 + \frac{1}{2}(dr^2 + r^2|de|^2).$$

Since $G = \rho^2 K(e)$, the adaptive metric is uniformly equivalent to

$$\frac{|dc|^2 + dr^2}{r^2} + |de|^2, \quad (15)$$

the product of an upper-half-space hyperbolic metric with the round sphere. This model is complete, and uniform equivalence preserves metric completeness. Hopf–Rinow finishes the proof. $\square$

Theorem 9 (Exact scalar adaptive completeness criterion). *Let $\Phi(z) = \varphi(|z|)I_d$ be a continuous strictly positive-definite scalar radial profile, C^2 away from the origin. Put*

$$\delta(r) = \varphi(0) - \varphi(r),$$

and suppose δ is nondecreasing on some interval $(0, r_0)$. For the RMS-normalized adaptive power family:

1. *if $N = 2$, the metric is geodesically complete exactly when $\alpha = 2$;*
2. *if $N \geq 3$, the metric is geodesically complete exactly when*

$$\alpha = 2 \quad \text{and} \quad \int_0^{r_0} \frac{dr}{\sqrt{\delta(r)}} = \infty. \quad (16)$$

When the integral is finite, incompleteness is witnessed by a finite-length curve along which exactly two landmarks collide while ρ stays bounded away from zero.

Proof. The $N = 2$ statement is Theorem 8. Let $N \geq 3$. If the integral diverges, define

$$\omega(t)^2 = 2 \sup_{0 < s \leq t} \delta(s).$$

Two-point strict positive definiteness gives $\delta(s) > 0$ for $s > 0$, and the kernel Cauchy–Schwarz inequality bounds δ , so ω is positive and nondecreasing. Near zero, $\omega(t)^2 = 2\delta(t)$, and the scalar feature gap is

$$2\Phi(0) - \Phi(z) - \Phi(-z) = 2\delta(|z|)I_d \preceq \omega(|z|)^2 I_d.$$

Theorem 6 proves completeness at $\alpha = 2$; Theorem 4 excludes every other exponent.

Conversely, suppose the integral in (16) is finite. Monotonicity forces

$$\frac{\delta(r)}{r^2} \longrightarrow \infty \quad (r \downarrow 0). \quad (17)$$

Indeed, otherwise there are $C < \infty$ and a sequence $r_j \downarrow 0$, chosen with $r_{j+1} < r_j/2$, such that $\delta(r_j) \leq Cr_j^2$. On every disjoint interval $[r_j/2, r_j]$, monotonicity would give

$$\int_{r_j/2}^{r_j} \frac{dr}{\sqrt{\delta(r)}} \geq \frac{1}{2\sqrt{C}},$$

contradicting finiteness of the integral.

Fix a unit vector e , choose mutually distinct nonzero sites $x_3, \dots, x_N$, and consider

$$q^1(\varepsilon) = -\varepsilon e, \quad q^2(\varepsilon) = \varepsilon e, \quad q^j(\varepsilon) = x_j \quad (j \geq 3). \quad (18)$$

The mean is independent of ε , while

$$\rho(\varepsilon)^2 = \rho(0)^2 + \frac{2}{N}\varepsilon^2, \quad \rho(0) > 0.$$

Set $t_\varepsilon = 2\varepsilon/\rho(\varepsilon)$, and let K_ε be the normalized scalar Gram matrix. In the basis whose first vector is

$$u = \frac{1}{\sqrt{2}}(1, -1, 0, \dots, 0),$$

write

$$K_\varepsilon = \begin{pmatrix} a_\varepsilon & b_\varepsilon^\top \\ b_\varepsilon & C_\varepsilon \end{pmatrix}, \quad a_\varepsilon = \delta(t_\varepsilon).$$

At $\varepsilon = 0$, C_0 is the weighted Gram matrix of the $N - 1$ distinct normalized sites

$$0, \quad x_3/\rho(0), \quad \dots, \quad x_N/\rho(0),$$

and hence is positive definite. Smoothness at their nonzero separations yields

$$b_\varepsilon = O(\varepsilon), \quad C_\varepsilon^{-1} = O(1).$$

Since $t_\varepsilon \sim 2\varepsilon/\rho(0)$, (17) gives $\varepsilon^2 = o(a_\varepsilon)$. Thus the Schur complement satisfies

$$S_\varepsilon = a_\varepsilon - b_\varepsilon^\top C_\varepsilon^{-1} b_\varepsilon = a_\varepsilon(1 + o(1)). \quad (19)$$

The path velocity is $-\sqrt{2}u \otimes e$, and exact block inversion gives

$$|\dot{q}(\varepsilon)|_g^2 = \frac{2}{\rho(\varepsilon)^2 S_\varepsilon}.$$

The map $\varepsilon \mapsto t_\varepsilon$ is a C^1 diffeomorphism near zero with derivative bounded above and below. Consequently the remaining path length is bounded by a constant times

$$\int_0^{\varepsilon_0} \frac{d\varepsilon}{\sqrt{\delta(t_\varepsilon)}} \asymp \int_0^{t_{\varepsilon_0}} \frac{dt}{\sqrt{\delta(t)}} < \infty.$$

Its limit has $q^1 = q^2$, so the metric is incomplete. □

Corollary 5 (Exact power-exponential phase diagram). *For $0 < \beta \leq 2$, let*

$$\Phi_\beta(z) = e^{-|z|^\beta} I_d.$$

The corresponding RMS-normalized adaptive metric is geodesically complete exactly for

<i>number of landmarks</i>	<i>complete parameters</i>
$N = 2$	$\alpha = 2, \quad 0 < \beta \leq 2,$
$N \geq 3$	$\alpha = 2, \quad \beta = 2.$

Proof. Schoenberg's theorem [8] gives strict positive definiteness: $t \mapsto e^{-t^{\beta/2}}$ is a positive mixture of Gaussian profiles with positive scales, each of which is strict. Moreover $\delta(r) = 1 - e^{-r^\beta}$ is increasing and $\delta(r) \sim r^\beta$. Hence $\int_0 dr / \sqrt{\delta(r)}$ diverges exactly when $\beta \geq 2$. Apply Theorem 9. $\square$

Theorem 10 (Exact matrix-TRI adaptive completeness criterion). *Let Φ be a continuous strictly positive-definite TRI matrix profile, C^2 away from the origin, and write*

$$\Phi(0) = aI_d, \quad \Phi(re) = k_\parallel(r)P_e + k_\perp(r)P_e^\perp \quad (r > 0, |e| = 1).$$

Put

$$\delta_\parallel(r) = a - k_\parallel(r),$$

and suppose $\delta_\parallel$ is nondecreasing on some interval $(0, r_0)$. For the RMS-normalized adaptive power family:

1. *if $N = 2$, the metric is geodesically complete exactly when $\alpha = 2$;*
2. *if $N \geq 3$, the metric is geodesically complete exactly when*

$$\alpha = 2 \quad \text{and} \quad \int_0^{r_0} \frac{dr}{\sqrt{\delta_\parallel(r)}} = \infty. \quad (20)$$

Thus collision completeness depends only on the longitudinal gap; no condition on the tangential gap $\delta_\perp(r) = a - k_\perp(r)$ is needed.

Proof. The two-landmark statement is Theorem 8. Let $N \geq 3$. For $z = te$, the directional feature gap in (14) is

$$e^\top [2\Phi(0) - \Phi(te) - \Phi(-te)]e = 2\delta_\parallel(t).$$

If the integral in (20) diverges, take

$$\omega(t)^2 = 2 \sup_{0 < s \leq t} \delta_\parallel(s).$$

Near zero this is $2\delta_\parallel(t)$, and Theorem 7 proves completeness at $\alpha = 2$. Theorem 4 excludes the other exponents.

Conversely, suppose the integral is finite. The dyadic argument proving (17) applies verbatim and gives

$$\frac{\delta_\parallel(r)}{r^2} \longrightarrow \infty.$$

Fix a unit vector e , choose distinct nonzero scalars $x_3, \dots, x_N$, and use the collinear path

$$q^1 = -\varepsilon e, \quad q^2 = \varepsilon e, \quad q^j = x_j e \quad (j \geq 3).$$

All normalized differences are parallel to e , so every Gram block preserves $\mathbb{R}e \oplus e^\perp$. The collision velocity lies in the longitudinal sector. On that sector the normalized Gram matrix is the scalar matrix

$$(K_\varepsilon^\parallel)_{ij} = k_\parallel \left(\frac{|q^i - q^j|}{\rho(\varepsilon)} \right).$$

Strict matrix positive definiteness implies strict positive definiteness of this scalar Gram matrix: test the original kernel only on coefficients parallel to e .

In the landmark basis whose first vector is $u = (1, -1, 0, \dots, 0)/\sqrt{2}$, write

$$K_\varepsilon^\parallel = \begin{pmatrix} a_\varepsilon & b_\varepsilon^\top \\ b_\varepsilon & C_\varepsilon \end{pmatrix}.$$

Exactly as in the scalar proof,

$$a_\varepsilon = \delta_\parallel(2\varepsilon/\rho(\varepsilon)), \quad b_\varepsilon = O(\varepsilon), \quad C_\varepsilon^{-1} = O(1).$$

The superquadratic ratio makes $\varepsilon^2 = o(a_\varepsilon)$, hence

$$S_\varepsilon^\parallel = a_\varepsilon - b_\varepsilon^\top C_\varepsilon^{-1} b_\varepsilon = a_\varepsilon(1 + o(1)).$$

Exact block inversion in the invariant longitudinal sector gives

$$|\dot{q}(\varepsilon)|_g^2 = \frac{2}{\rho(\varepsilon)^2 S_\varepsilon^\parallel}.$$

The same C^1 change of variable $t = 2\varepsilon/\rho(\varepsilon)$ as in Theorem 9 shows that the collision path has finite length. Its limit has $q^1 = q^2$, proving incompleteness. $\square$

Corollary 6 (Genuinely matrix-valued power-exponential phase diagram). *Let $d \geq 2$, $\eta > 0$, $0 < \beta \leq 2$, and*

$$\Phi_{\beta,\eta}(z) = e^{-|z|^\beta} I_d + \eta \left[-D^2 e^{-|z|^\beta} \right].$$

This is a strict, genuinely matrix-valued TRI kernel. Its adaptive metric is complete exactly for

<i>number of landmarks</i>	<i>complete parameters</i>
$N = 2$	$\alpha = 2, \quad 0 < \beta \leq 2,$
$N \geq 3$	$\alpha = 2, \quad \beta = 2.$

Proof. Schoenberg's theorem [8] gives strict positive definiteness of the first summand. The Fourier transform of the second is a positive scalar multiple of $\xi \xi^\top e^{-|\xi|^2/4}$, so it is a positive-semidefinite matrix kernel. Their sum is strict, and for $\eta > 0$ it is not a scalar multiple of I_d .

Along $z = re$, its longitudinal coefficient is

$$k_\parallel(r) = e^{-r^\beta} + \eta(2 - 4r^2)e^{-r^2},$$

while $\Phi_{\beta,\eta}(0) = (1 + 2\eta)I_d$. Hence

$$\delta_\parallel(r) = 1 - e^{-r^\beta} + \eta \left[2 - (2 - 4r^2)e^{-r^2} \right].$$

This is increasing near zero and is asymptotic to r^β for $\beta < 2$, and to $(1 + 6\eta)r^2$ for $\beta = 2$. Apply Theorem 10. $\square$

Theorem 11 (Exact full two-landmark scalar geometry). *Let $\Phi(z) = \varphi(|z|)I_d$ be any continuous strictly positive-definite scalar radial profile. For the adaptive construction with $N = 2$ and $\alpha = 2$, set*

$$a = \varphi(0), \quad b = \varphi(2), \quad B = \frac{2}{a - b}.$$

Write

$$c = \frac{q^1 + q^2}{2}, \quad q^1 - q^2 = r\omega, \quad x = 2\sqrt{\frac{a - b}{a + b}}c.$$

Then $a \pm b > 0$, and the entire two-landmark metric is

$$g = B \left[\frac{dr^2 + |dx|^2}{r^2} + g_{S^{d-1}} \right]. \quad (21)$$

Thus each connected component is isometric to the scaled product

$$B \left(\mathbb{H}^{d+1} \times S^{d-1} \right).$$

For the corresponding adaptive power family, the full two-landmark metric is complete if and only if $\alpha = 2$.

Proof. Use the canonical variables

$$z = q^1 - q^2, \quad P = p_1 + p_2, \quad \pi = \frac{p_1 - p_2}{2}.$$

Since $\rho = r/2$, the diagonal and off-diagonal cometric blocks are $\rho^2 a I_d$ and $\rho^2 b I_d$. The Hamiltonian is

$$H = \frac{\rho^2(a + b)}{4} |P|^2 + \rho^2(a - b) |\pi|^2.$$

Strict two-point positive definiteness gives $a \pm b > 0$. Inverting the center and relative cometric blocks gives

$$g = \frac{8}{r^2(a + b)} |dc|^2 + \frac{2}{a - b} \left(\frac{dr^2}{r^2} + g_{S^{d-1}} \right).$$

The stated rescaling of c turns this into (21); the first term in brackets is the upper-half-space metric on $\mathbb{H}^{d+1}$. The product is complete. All other exponents are incomplete by Theorem 4. $\square$

Corollary 7 (Distance, curvature, cut locus, and conjugate locus). *Represent a point in Theorem 11 by (h, ω) , where $h = (x, r) \in \mathbb{H}^{d+1}$. For $d \geq 2$,*

$$\begin{aligned} d_g((h_0, \omega_0), (h_1, \omega_1))^2 &= B \left[d_{\mathbb{H}}(h_0, h_1)^2 + \theta^2 \right], \\ \theta &= \arccos(\omega_0^\top \omega_1), \\ \cosh d_{\mathbb{H}}(h_0, h_1) &= 1 + \frac{|x_0 - x_1|^2 + (r_0 - r_1)^2}{2r_0 r_1}. \end{aligned}$$

Sectional curvature is $-1/B$ on hyperbolic planes, $1/B$ on spherical planes, and zero on mixed planes. The injectivity radius is $\pi\sqrt{B}$, and the cut locus of (h_0, ω_0) is

$$\mathbb{H}^{d+1} \times \{-\omega_0\}.$$

For $d \geq 3$, the exponential is singular exactly at tangent vectors whose standard-sphere component has length $m\pi$, $m = 1, 2, \dots$, with multiplicity $d - 2$. The first conjugate locus is $\mathbb{H}^{d+1} \times \{-\omega_0\}$, while the set of all conjugate endpoints is

$$\mathbb{H}^{d+1} \times \{\omega_0, -\omega_0\}.$$

In particular, a pure angular great circle has its first ambient conjugate point at length $\pi\sqrt{B}$, with multiplicity $d - 2$. For $d = 2$, the circle factor has cut points but no conjugate points, so the full manifold has no conjugate points. For $d = 1$, each order component is a scaled hyperbolic plane.

Proof. All assertions are the standard product consequences of (21). Hyperbolic space has no cut or conjugate points. On the unit round sphere, the cut point is the antipode and the differential of the exponential is singular when the tangent length is $m\pi$, with kernel dimension $d - 2$. Multiplying the metric by B multiplies lengths by $\sqrt{B}$ and curvatures by $1/B$. $\square$

Corollary 8 (Adaptive matrix-TRI fixed-center product). *Let Φ be a strictly positive-definite TRI matrix profile and write*

$$\Phi(0) = aI_d, \quad \Phi(2\omega) = k_{\parallel}(2)P_{\parallel} + k_{\perp}(2)P_{\perp}.$$

Set

$$\delta_{\parallel} = a - k_{\parallel}(2), \quad \delta_{\perp} = a - k_{\perp}(2).$$

For $N = 2$ and $\alpha = 2$, the fixed-center sector is totally geodesic and has the exact product metric

$$g_{\text{rel}} = \frac{2}{\delta_{\parallel}} ds^2 + \frac{2}{\delta_{\perp}} g_{S^{d-1}}, \quad s = \log |q^1 - q^2|. \quad (22)$$

Its distance, cut locus, and conjugate locus are the corresponding line-sphere product formulas. For $d \geq 3$, the first pure-angular sector conjugate length is

$$\pi\sqrt{\frac{2}{\delta_{\perp}}},$$

with multiplicity $d - 2$; these sector Jacobi fields are also ambient Jacobi fields.

Proof. Translation invariance conserves $P = p_1 + p_2$, and the center velocity vanishes when $P = 0$, so the fixed-center sector is totally geodesic. The relative cometric is

$$2\rho^2[\Phi(0) - \Phi(2\omega)] = \frac{r^2}{2}[\delta_{\parallel}P_{\parallel} + \delta_{\perp}P_{\perp}].$$

Inversion and $z = r\omega$ give (22). The remaining claims follow from the product exponential and Jacobi equation. $\square$

Regularity and Generic Two-Landmark TRI Geometry

For fixed scalar radial kernels, the global completeness question is already answered. Bauer–Bruveris–Michor’s Theorem 9.8 gives completeness under the standard C_b^1 -admissibility hypothesis [5], and Habermann–Preston–Sommer prove the sharp all- N , all- d criterion

$$\int_0^a \frac{dr}{\sqrt{K(0) - K(r)}} = \infty$$

for scalar radial kernels smooth away from the origin [6]. The next theorem records the matrix-valued two-landmark reduction suggested by the source.

Theorem 12 (Matrix-TRI relative reduction). *Let κ be a continuous, strictly positive-definite, rotation- and translation-invariant matrix kernel, C^2 away from the origin. Write*

$$\kappa(0) = aI_d, \quad \kappa(z) = k_{\parallel}(r)P_{\parallel} + k_{\perp}(r)P_{\perp}, \quad r = |z|,$$

and set $\delta_{\parallel} = a - k_{\parallel}$, $\delta_{\perp} = a - k_{\perp}$. On the two-landmark fixed-center sector,

$$g_{\text{rel}} = \frac{dr^2}{2\delta_{\parallel}(r)} + \frac{r^2}{2\delta_{\perp}(r)} g_{S^{d-1}}. \quad (23)$$

It is complete if and only if

$$\int_0^\epsilon \frac{dr}{\sqrt{\delta_{\parallel}(r)}} = \infty.$$

If

$$\frac{ds}{dr} = \frac{1}{\sqrt{2\delta_{\parallel}(r)}}, \quad f(s) = \frac{r}{\sqrt{2\delta_{\perp}(r)}},$$

then its radial and tangential sectional curvatures are

$$K_{\text{rad}} = -\frac{f''(s)}{f(s)}, \quad K_{\text{tan}} = \frac{1 - f'(s)^2}{f(s)^2} \quad (d \geq 3).$$

Proof. In the canonical center/relative variables used above, setting $P = 0$ gives

$$H_{\text{rel}} = \pi^T(\kappa(0) - \kappa(z))\pi.$$

The radial and tangential eigenvalues of the relative cometric are therefore $2\delta_{\parallel}$ and $2\delta_{\perp}$. Inversion proves (23). Any curve changing radius from r_0 to r_1 has length at least the integral of $dr/\sqrt{2\delta_{\parallel}(r)}$, with equality for a radial curve. Two-point positive definiteness gives $0 < \delta_{\parallel}(r) \leq 2a$, so Euclidean infinity is always infinitely far away. On compact radial annuli the metric is uniformly nondegenerate; the displayed collision integral is therefore necessary and sufficient for completeness. The change to s turns (23) into the warped product $ds^2 + f(s)^2 g_{S^{d-1}}$, whose standard curvature formulas give the last assertion. $\square$

For a scalar kernel $\delta_{\parallel} = \delta_{\perp} = \delta$, the relative Gaussian curvature in $d = 2$ is

$$K_{\text{rel}}(r) = \delta(r) \left(\frac{d^2}{dr^2} + \frac{1}{r} \frac{d}{dr} \right) \log \delta(r).$$

For $K(r) = e^{-r^2/(2\sigma^2)}$, with $t = r^2/(2\sigma^2)$, this becomes

$$K_{\text{rel}}(r) = \frac{2}{\sigma^2} \frac{e^t(1-t) - 1}{e^t(e^t - 1)} < 0.$$

This intrinsic fixed-center surface is complete and has no conjugate points. The broader scalar landmark curvature and conjugate-point mechanisms were already developed by Micheli–Michor–Mumford [7]; this calculation is included to connect Theorem 12 to that literature, not as a new scalar result.

What Kernel Tails Control

Theorem 13 (Quantitative cluster decoupling). *Let κ be a stationary C^2 matrix kernel and define*

$$\eta(R) = \sup_{|x| \geq R} (\|\kappa(x)\| + \|D\kappa(x)\|).$$

Partition the landmarks into clusters and let X_0 be the Hamiltonian system obtained by deleting cross-cluster terms. On a phase-space region where $|p_a| \leq P$ and every intercluster separation is at least R , the full and decoupled vector fields satisfy

$$\|F - F_0\| \leq C(N, P)\eta(R).$$

If F_0 has Lipschitz constant L on that region and both trajectories remain there through time T , then equal initial data satisfy

$$\|X(t) - X_0(t)\| \leq C(N, P)\eta(R) \frac{e^{Lt} - 1}{L}, \quad 0 \leq t \leq T,$$

with the usual interpretation t when $L = 0$.

Proof. Every omitted term in a position equation is $\kappa(q^a - q^b)p_b$, bounded by $P\eta(R)$. Every omitted term in a momentum equation is bilinear in p_a, p_b and contains $D\kappa(q^a - q^b)$, hence is bounded by $P^2\eta(R)$. Summing at most N terms per component gives the first estimate. Write

$$F(X) - F_0(X_0) = [F(X) - F_0(X)] + [F_0(X) - F_0(X_0)]$$

and apply Grönwall's inequality to obtain the trajectory bound. $\square$

Thus tail decay has a universal mathematical meaning: it controls the error made by treating distant landmark clusters as independent. It cannot by itself decide which kernel is statistically optimal for a particular data analysis task.

The Full Arbitrary-Matrix Collision-End Reduction

Theorem 14 (Exact normalized-shape completeness criterion). *Let $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ be continuous and C^2 away from the origin, and suppose its stationary matrix kernel is strictly positive definite. Define the normalized pre-shape manifold*

$$\Sigma_N = \left\{ x \in (\mathbb{R}^d)^N : \sum_a x^a = 0, \quad \frac{1}{N} \sum_a |x^a|^2 = 1, \quad x^a \neq x^b \ (a \neq b) \right\}.$$

For $x \in \Sigma_N$, put

$$K(x)_{ab} = \Phi(x^a - x^b), \quad A(x) = K(x)^{-1}.$$

If $v \in T_x \Sigma_N$, set

$$h_x(v, v) = \min_{\substack{u \in \mathbb{R}^d \\ \lambda \in \mathbb{R}}} (v + \mathbf{1}_N \otimes u + \lambda x)^\top A(x) (v + \mathbf{1}_N \otimes u + \lambda x). \quad (24)$$

Then h is a Riemannian metric. For every $N \geq 2$, the RMS-normalized adaptive power metric is geodesically complete on every component of $\mathcal{Q}_N$ if and only if

$\alpha = 2 \quad \text{and} \quad (\Sigma_N, h) \text{ is complete on every component.}$

Equivalently, because the closure of Σ_N is compact, every deleted collision diagonal must lie at infinite h -distance.

More explicitly, let

$$V_x : \mathbb{R}^{d+1} \longrightarrow (\mathbb{R}^d)^N, \quad V_x(u, \lambda) = \mathbf{1}_N \otimes u + \lambda x.$$

Then

$$h_x(v, v) = v^\top \left[A - AV_x(V_x^\top AV_x)^{-1}V_x^\top A \right] v. \quad (25)$$

Thus the criterion is an explicit finite-dimensional Schur-complement test on a compact normalized shape space, with no scalar, radial, TRI, monotonicity, or regular-variation hypothesis.

Proof. Every configuration has the unique decomposition

$$q = \bar{q} \mathbf{1}_N + \rho(q)x, \quad x \in \Sigma_N.$$

At $\alpha = 2$, translations and positive dilations act isometrically. Their tangent space at the representative x is

$$\mathcal{V}_x = \{\mathbf{1}_N \otimes u + \lambda x : u \in \mathbb{R}^d, \lambda \in \mathbb{R}\}.$$

The metric at x is $A(x)$. Formula (24) is therefore exactly the Riemannian quotient metric. The columns of V_x are independent, and ordinary weighted least squares gives (25). Strict positive definiteness of $K(x)$ makes the result positive on $T_x \Sigma_N$.

Suppose first that a curve in $\mathcal{Q}_N$ has finite g -length. The scale estimate (10) keeps ρ bounded above and away from zero. The operator estimate (9) then makes the curve Euclidean Cauchy. Hence it has a finite Euclidean limit, and the only way that limit can leave $\mathcal{Q}_N$ is through a collision. Its projection to Σ_N has no greater length, because quotient maps decrease length. Consequently completeness of h excludes every finite-length escape of the original metric.

Conversely, suppose h is incomplete. Since Σ_N is an open submanifold of a compact sphere, there is a finite-length piecewise C^1 curve $x(t)$, $0 \leq t < 1$, approaching a collision diagonal. Take its horizontal lift for the translation–dilation quotient. On every compact subinterval this lift exists and has exactly the same length. It cannot cease at a time $t < 1$: the scale and Euclidean estimates just used give a finite noncolliding limit because $x(t)$ is still in Σ_N , so the lift extends. The complete lift on $[0, 1)$ has finite length. The same estimates give a positive finite limiting scale and a finite Euclidean limit, while its normalized shape tends to a collision diagonal. Its limit therefore lies outside $\mathcal{Q}_N$, proving incompleteness.

For $\alpha \neq 2$, Theorem 4 gives a finite-length radial escape. This proves the equivalence. $\square$

Remark 4. For $N = 2$, $\Sigma_2 = S^{d-1}$: normalization freezes the two landmarks at antipodal points and there is no deleted collision diagonal. Theorem 14 therefore recovers universal matrix two-landmark completeness. For $N \geq 3$, Theorem 7 is a profile-only sufficient certificate that all quotient collision ends are infinitely far away. Theorems 9 and 10 are the cases in which a quotient end reduces further to one scalar integral.

Theorem 15 (Full arbitrary-matrix two-landmark block law). *Let $N = 2$, $\alpha = 2$, and write*

$$q^1 = c + \frac{z}{2}, \quad q^2 = c - \frac{z}{2}, \quad z = re, \quad |e| = 1.$$

Set

$$A = \Phi(0), \quad B(e) = \Phi(2e), \quad B_s = \frac{B + B^\top}{2}, \quad C = \frac{B - B^\top}{2},$$

and define

$$\mathcal{J}(e) = \begin{pmatrix} (A + B_s)/2 & -C \\ C & 2(A - B_s) \end{pmatrix}. \quad (26)$$

Then $\mathcal{J}(e)$ is positive definite and the full two-landmark metric is

$$g = \frac{4}{r^2} \begin{pmatrix} dc \\ dz \end{pmatrix}^\top \mathcal{J}(e)^{-1} \begin{pmatrix} dc \\ dz \end{pmatrix}. \quad (27)$$

In the canonical momenta

$$P = p_1 + p_2, \quad \pi = \frac{p_1 - p_2}{2},$$

the zero-total-momentum sector $P = 0$ has center drift

$$\dot{c} = -\rho^2 C(e) \pi. \quad (28)$$

Thus the odd skew part of a general stationary matrix profile is the exact obstruction to the fixed-center reduction.

Proof. The normalized two-point Gram matrix is

$$K(e) = \begin{pmatrix} A & B \\ B^\top & A \end{pmatrix}.$$

Substitute

$$p_1 = P/2 + \pi, \quad p_2 = P/2 - \pi$$

into $p^\top K(e) p$. Direct expansion gives

$$p^\top K(e) p = \begin{pmatrix} P \\ \pi \end{pmatrix}^\top \mathcal{J}(e) \begin{pmatrix} P \\ \pi \end{pmatrix}.$$

Hence $\mathcal{J}$ is a congruence transform of the strictly positive matrix K . Since $\rho = r/2$, inversion gives (27). Hamilton's center equation is

$$\dot{c} = \rho^2 \left[\frac{A + B_s}{2} P - C\pi \right],$$

which gives (28). □

Corollary 9 (Even profiles give an exact magnetic sphere reduction). *In Theorem 15, suppose $\Phi(z) = \Phi(-z)$. Then $\Phi(z)$ is symmetric, $C = 0$, and the fixed-center sector is totally geodesic. Put*

$$D(e) = A - \Phi(2e), \quad H(e) = D(e)^{-1}, \quad s = \log r.$$

Its relative metric has the exact Kaluza–Klein form

$$g_{\text{rel}} = 2(e \, ds + de)^\top H(e) (e \, ds + de) = a(e)(ds + \theta_e)^2 + h_e, \quad (29)$$

where, for $de \in T_e S^{d-1}$,

$$\begin{aligned} a(e) &= 2e^\top H(e)e, \\ \theta_e(de) &= \frac{e^\top H(e)de}{e^\top H(e)e}, \\ h_e(de, de) &= 2 \left[de^\top H(e)de - \frac{(e^\top H(e)de)^2}{e^\top H(e)e} \right]. \end{aligned}$$

If $J = a(e)(\dot{s} + \theta_e(\dot{e}))$ is the conserved logarithmic-scale momentum, the direction $e(t)$ is governed by the Routhian

$$L_J(e, \dot{e}) = \frac{1}{2}h_e(\dot{e}, \dot{e}) + J\theta_e(\dot{e}) - \frac{J^2}{2a(e)}, \quad (30)$$

and

$$\dot{s} = \frac{J}{a(e)} - \theta_e(\dot{e}).$$

Thus relative geodesics and their Jacobi equations reduce exactly to magnetic mechanics on the compact direction sphere, with magnetic field $J d\theta$ and potential $J^2/(2a)$.

For a TRI profile,

$$a = \frac{2}{\delta_{\parallel}}, \quad \theta = 0, \quad h = \frac{2}{\delta_{\perp}}g_{S^{d-1}},$$

so (29) reduces to the line–sphere product in Corollary 8.

Proof. Evenness and stationarity give $\Phi(z) = \Phi(-z) = \Phi(z)^{\top}$. Formula (26) is block diagonal. Translation invariance conserves P , and $\dot{c} = 0$ is then equivalent to $P = 0$, proving that the fixed-center sector is totally geodesic. Its relative cometric is $2\rho^2 D(e)$. Inverting, using $\rho = r/2$, and substituting $dz = r(e ds + de)$ gives the first expression in (29). Completing the square gives a, θ, h .

For the Lagrangian

$$L = \frac{1}{2}a(e)(\dot{s} + \theta_e(\dot{e}))^2 + \frac{1}{2}h_e(\dot{e}, \dot{e}),$$

the cyclic momentum is $J = a(\dot{s} + \theta(\dot{e}))$. Eliminating $\dot{s}$ from $L - J\dot{s}$ gives (30), and reconstruction gives the displayed equation for $\dot{s}$. The TRI specialization follows by diagonalizing $D(e)$ on $\mathbb{R}e \oplus e^{\perp}$. $\square$

Verification

The accompanying script `code/verify_multiscale_covariance.py` differentiates the scalar Gaussian Hamiltonians explicitly. It checks:

1. all four augmented Hamilton equations for $N = 3$;
2. the adaptive RMS-bandwidth q - and p -equations for $N = 3$;
3. five exponents $\alpha \in \{-1.5, 0, 1, 2, 3.25\}$;
4. three dilation factors $\lambda \in \{0.4, 1.8, 3\}$;
5. positive Gaussian Gram eigenvalues.

All 30 covariance cases pass with relative and absolute tolerances $2 \cdot 10^{-13}$. This is a sanity check, not a substitute for the analytic proofs.

The second script, `code/verify_full_geometry.py`, checks the linear-algebra identities for the Gaussian specialization of Theorem 5 on 12 deterministic random configurations with $N \in \{2, 3, 5, 8\}$ and $d \in \{1, 2, 3\}$. It verifies positive Gram eigenvalues, the operator and $d\rho$ bounds, all 126 pair-cometric inequalities, and degree-two homogeneity. It also checks the exact two-landmark product constant and Jacobi zero, negativity of the stationary relative Gaussian curvature on 80 logarithmically spaced radii, and the radial exponent thresholds. Every check passes at tolerance $3 \cdot 10^{-12}$. These computations test formulas and signs; completeness and conjugacy are proved analytically above.

The third script, `code/verify_admissible_profile_geometry.py`, uses a strict non-TRI matrix-valued Gaussian mixture on all 16 combinations

$$N \in \{2, 3, 5, 8\}, \quad d \in \{1, 2, 3, 4\}.$$

It verifies the block trace and $d\rho$ bounds, all 168 pair-feature inequalities, and exact degree-two homogeneity. It then checks 80 random tangent identities for the full scalar $\mathbb{H}^{d+1} \times S^{d-1}$ metric and 60 identities for the fixed-center product of a strict matrix-TRI mixture containing a curl-free Gaussian component. It terminates with **ALL ADMISSIBLE-PROFILE GEOMETRY CHECKS PASSED**.

The fourth script is stored as

`code/verify_osgood_phase_diagram.py`.

It performs 80-decimal Schur-complement checks for all 16 combinations

$$(N, d) \in \{(3, 1), (4, 2), (6, 3), (8, 4)\}, \quad \beta \in \{0.5, 1, 1.5, 1.9\}.$$

At collision scales 10^{-12} and 10^{-18} , it verifies the exact inverse/Schur identity, $S_\varepsilon/a_\varepsilon \rightarrow 1$, the logarithmic slope $d \log S/d \log \varepsilon \rightarrow \beta$, the finite scaled speed, and the model integral threshold. It terminates with **ALL OSGOOD PHASE-DIAGRAM CHECKS PASSED**. This is a high-precision stress test of (19); the asymptotic proof is analytic.

The fifth script is

`code/verify_matrix_tri_phase_diagram.py`.

It uses 180-decimal arithmetic for all 16 combinations

$$(N, d) \in \{(3, 2), (4, 3), (6, 4), (8, 5)\}, \quad \beta \in \{0.5, 1, 1.5, 1.9\},$$

for the genuinely matrix-valued profile in Corollary 6. At collision scales 10^{-20} and 10^{-50} , it verifies the full matrix inverse against the longitudinal Schur formula, the exact antisymmetric-block/gap identity, $S_\varepsilon^\parallel/\delta_\parallel \rightarrow 1$, the logarithmic exponent, and finite scaled collision speeds. It also checks the quadratic logarithmic threshold and 240 tangent comparisons for Theorem 8 using a strict non-TRI Gaussian matrix mixture. It terminates with **ALL MATRIX-TRI PHASE-DIAGRAM CHECKS PASSED**.

The sixth script is

`code/verify_shape_quotient_and_general_n2.py`.

For strict non-TRI Gaussian matrix mixtures it checks 100 identities: Schur complements, horizontal orthogonality, horizontal lifts, and quotient isometries on

$$(N, d) \in \{(3, 2), (4, 2), (5, 3), (7, 3)\}.$$

It then uses a strict real stationary spectral Gaussian kernel with a nonzero odd skew part to check 160 full center-relative cotangent and tangent block identities and the exact center drift (28). Finally it checks 180 Kaluza–Klein completions (29) for strict even non-TRI profiles. It terminates with **ALL SHAPE-QUOTIENT AND GENERAL N=2 CHECKS PASSED**.

Literature Boundary, Limitations, and Review

Niethammer–Vialard [2] substantially anticipates the negative fixed-LDDMM conclusion and explicitly says one must go beyond standard LDDMM to obtain scale invariance. Their positive construction separates a global volume coordinate from unit-volume shape and characterizes preservation of that indicator under parallel transport. The present adaptive construction instead gives a direct homogeneous cometric on landmark configurations, and Theorem 3 classifies a natural dynamical global-scale kernel ansatz.

Multiscale LDDMM itself is established literature. The kernel-bundle work of Sommer et al. [3] combines deformation at multiple scales, while Liu–Younes [4] construct an RKHS of vector fields over a continuous scale interval. Fixed scalar-kernel completeness is also now completely settled by Habermann–Preston–Sommer [6], building on the admissible kernel result in Bauer–Bruveris–Michor [5]. Scalar landmark curvature and examples of conjugate points predate the source paper in Micheli–Michor–Mumford [7]. None of these literature results is claimed as new here.

The fixed-kernel result of Habermann–Preston–Sommer is especially close in mechanism to Theorems 6 and 9: their criterion is an integral test involving the scalar kernel gap at a collision, and for a gap of order r^γ it has the same threshold $\gamma = 2$. Their cometric is nevertheless generated by one fixed stationary kernel. Here the bandwidth is the global configuration scale $\rho(q)$; this removes the binary shape-collision boundary entirely when $N = 2$, but leaves a sharp embedded binary collision for $N \geq 3$.

The bounded searches performed on 2026-07-22 and 2026-07-23 used arXiv ids 1308.5739, 1009.2637, 1305.1150, 2503.10611, and 2501.04031, together with the phrases “scale-invariant landmark metric”, “homogeneous cometric”, “configuration-dependent bandwidth”, “matrix-valued TRI landmark completeness”, “adaptive landmark kernel”, “landmark hyperbolic sphere product”, “two-landmark hyperbolic geometry”, “adaptive cometric landmark”, “landmark cut locus”, “landmark kernel tail decoupling”, “Osgood landmark completeness”, “RMS-normalized landmark kernel”, “power-exponential landmark metric”, and “rough adaptive landmark kernel”. The final matrix pass added “TRI kernel completeness”, “longitudinal landmark kernel completeness”, “radial tangential matrix kernel landmark completeness”, “adaptive matrix-valued landmark metric”, and “two-landmark matrix-valued kernel completeness”. The arbitrary-matrix pass further used “normalized pre-shape sphere”, “shape quotient landmark completeness”, “matrix-valued landmark Kaluza–Klein”, “magnetic geodesic matrix kernel”, and close variants. They found no exact statement of Theorem 1, the power-profile iff law (7), Theorems 4, 5, 6, 7, 8, 9, 10, and 11, the arbitrary-matrix quotient classification in Theorem 14, or the general two-landmark block and magnetic reductions in Theorem 15 and Corollary 9, the Corollaries 5, 6, 7, and 8, the matrix-TRI warped reduction in Theorem 12, or the quantitative tail bound in Theorem 13. This is a bounded novelty assessment, not an exhaustive review.

Although its profile may come from an admissible RKHS, the adaptive metric is not induced by one fixed ambient kernel; its normalization depends on the whole configuration. The augmented model adds a global state variable. The packet gives an Osgood sufficient condition, an exact criterion for scalar monotone-gap profiles, and a sharp nonadmissible phase diagram for power-exponential profiles. The directional theorem and longitudinal Schur reduction now give an exact criterion for every matrix-TRI profile with a locally monotone longitudinal gap. Theorem 14 now supplies the exact necessary-and-sufficient criterion for every irregular non-TRI profile as well, but in its natural compact path-metric form rather than as a generally unavailable one-dimensional radial integral. The packet does not give a profile-independent closed formula for the full conjugate locus when $N \geq 3$, or solve the magnetic Jacobi equation in closed form for every arbitrary matrix profile. Nor can a geometry-only theorem select the statistically optimal scale functional for a particular

application. These are explicit boundaries, not hidden claims that standard LDDMM has become scale invariant.

Human review should focus on the quantifiers in the covariance definitions, the continuous-character step in Theorems 1 and 3, the differentiated scaling of the s -equation, the feature-map implication and logarithmic length estimates in Theorem 5, the Osgood reparametrization in Theorem 6, the radial-covector computation in Theorem 7, the compact-sphere uniform equivalence in Theorem 8, the monotone-gap necessity argument, limiting complementary Gram block, and Schur asymptotic in Theorem 9, the center/relative constants and hyperbolic coordinate change in Theorem 11, the product exponential in Corollary 7, and the separation of radial and tangential eigenvalues in Corollary 8 and Theorems 10 and 12, the horizontal-lift converse in Theorem 14, the skew signs in (26), and the Routhian sign in (30). The recommended status is *strong partial result, likely valid; exact scale covariance, all- N admissible and directional-Osgood adaptive completeness, exact arbitrary-matrix normalized-shape completeness, universal and structurally reduced two-landmark matrix geometry, exact scalar and matrix-TRI monotone-gap criteria and power-exponential phase diagrams, and the full scalar two-landmark ambient geometry complete in the stated classes.*

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